WEEK 05 TECHNICAL REPORT

YCB-V Error Analysis I

Intern	Mehrana Rezaei
Group	Group 1 — Basic ML Knowledge
Main Quest	Structuring Machine Learning Projects
Side Quest	Error Analysis I

1. Objective

The Week 05 objective was to diagnose, not retrain, the five-class YCB-V object classifier chosen in Week 04. The work re-examined the original E1_regularization evaluation, quantified every class-wise mistake and confusion transition, inspected the already-saved example figures, and organized plausible causes into a verification plan for Week 06.

The key question was why the same classifier reached 100.00% training accuracy and 98.93% validation accuracy but only 41.78% accuracy on held-out scenes. The analysis preserves the original BOP'19 subset and the exact Week 04 E1 confusion matrix; it does not introduce a new model, split, prediction set, or benchmark result.

2. Main Quest Progress and Relevant Concepts

The Main Quest for this week was Structuring Machine Learning Projects.The Week 05 notebook applies the evaluation, dataset-mismatch and human-guided error-analysis ideas where its available evidence allows; lesson coverage is not itself proof of a completed experiment.

- 01 — Why ML Strategy?
- 02 — Orthogonalization
- 03 — Single-Number Evaluation Metric
- 04 — Satisficing and Optimizing Metrics
- 05 — Train / Dev / Test Distributions
- 06 — Size of the Dev and Test Sets
- 07 — When to Change Dev/Test Sets and Metrics
- 08 — Why Human-Level Performance?
- 09 — Avoidable Bias
- 10 — Understanding Human-Level Performance
- 11 — Surpassing Human-Level Performance
- 12 — Improving Your Model’s Performance

2.1 Concepts directly connected to this Side Quest

- Train/dev/test distributions and set sizes: examined the 877 / 375 / 450 instance split and its 8 / 2 / 2 scene grouping; the large validation–test gap motivates a distribution/representativeness check.
- Single-number metrics versus diagnostic metrics: retained test accuracy for continuity, then broke performance into per-class precision, recall, F1, counts, and directed class confusions.
- Avoidable bias and error prioritization: near-perfect training fit does not, by itself, explain weak unseen-scene performance; the analysis separates observed metrics from hypotheses about scene shift and crop context.
- Human-level performance and orthogonalization: these lessons provide conceptual guidance for targeted manual review and distinct diagnostic steps; no human baseline was measured in this notebook.

3. Baseline, Scope, and Reproducibility

All statistics refer to the original Week 04 E1_regularization checkpoint and the internally held-out scenes from BOP'19 test-subset imagery. “Test” in this report denotes that internal scene split; these are not official YCB-V benchmark test-set scores. Week 05 reused the recorded outputs rather than loading a checkpoint and running fresh inference.

Setting	Original Week 04 configuration
Source	Lightweight YCB-V BOP'19 subset
Classes	tomato_soup_can (4); sugar_box (3); master_chef_can (1); tuna_fish_can (6); bleach_cleanser (12)
Usable-instance filter	Bounding box present; width and height ≥ 24 px; visible fraction ≥ 0.35
Input	Ground-truth bounding-box crop, 8% padding, RGB resize to 64 × 64; per-channel mean/std = 0.5 / 0.5
Split	877 train / 375 validation / 450 test instances; 8 / 2 / 2 scenes, no scene overlap
Classifier	Flattened RGB → 256 ReLU → dropout → 128 ReLU → dropout → five outputs
Selected run	E1_regularization: Adam, learning rate 1e−3, dropout 0.30, weight decay 1e−4; no augmentation
Checkpoint choice	Lowest validation loss; selected epoch 7, reported val loss 0.0327

4. Method

The executed notebook reconstructs the exact 5 × 5 E1 confusion matrix by transcribing the complete matrix in the original executed Week 04 notebook. It verifies the class supports [150, 75, 75, 75, 75], 450 test samples, 188 correct predictions, and 262 errors. From this existing evaluation it computes class-wise metrics, row-normalized error rates, eight nonzero misclassification transitions, and observed cross-split accuracy gaps.

Representative correct and incorrect predictions are carried forward from the original Week 04 executed figures. Optional per-scene accuracy, per-image confidence distributions, and systematic crop/label auditing require the original final_test_predictions.csv or saved E1 checkpoint plus BOP images. The executed Week 05 run did not find that CSV and kept checkpoint recovery disabled; those optional measurements therefore remain unreported.

5. Implementation

Notebook stage	What was executed / preserved	Purpose
Part A	Reused original E1 settings, full 5 × 5 matrix and verified totals	Establish auditable baseline
Part B	Displayed raw-count and row-normalized matrices	Separate absolute counts from class-specific error rates
Part C	Calculated precision, recall, F1, error contributions, all transitions, split gaps	Locate concentrated failures and compare evaluation splits
Part C2	Displayed original correct/incorrect Week 04 image grids	Qualitative evidence for manual review
Part D	Checked for original prediction CSV; file not found	Avoid fabricated scene/image/confidence statistics
Part E	Optional checkpoint inference disabled	Avoid unrequested retraining or unverified replacement results
Part F	Saved tables and issue hypotheses; manual review CSV had 0 rows	Prepare evidence-based follow-up

6. Results

6.1 Observed summary

Metric	Observed value	Interpretation
Correct / total test	188 / 450	262 test errors
Test accuracy	41.7778%	Original Week 04 E1 result; not a new Week 05 run
Macro F1	0.4224	Equal weight per class
Weighted F1	0.4263	Weighted by class support
Train accuracy	100.00%	E1 selected checkpoint
Validation accuracy	98.93%	E1 selected checkpoint

Metric	Observed value	Interpretation
Validation − test gap	57.15 percentage points	Large observed discrepancy
Week 03 → Week 04 test	51.33% → 41.78%	−9.55 percentage points; Week 03 value rounded

6.2 Exact class-wise performance

Class	N	Correct	Errors	Precision	Recall	F1
tomato_soup_can	150	74	76	0.4066	0.4933	0.4458
sugar_box	75	29	46	0.7073	0.3867	0.5000
master_chef_can	75	68	7	1.0000	0.9067	0.9510
tuna_fish_can	75	6	69	0.1053	0.0800	0.0909
bleach_cleanser	75	11	64	0.1078	0.1467	0.1243

Most test errors were associated with tomato_soup_can (76/262; 29.01%), tuna_fish_can (69/262; 26.34%), and bleach_cleanser (64/262; 24.43%). Master_chef_can had 68 correct predictions out of 75 and only seven errors. Class-specific recall makes the near-total tuna failure particularly clear: 6/75 = 8.00%.

6.3 Full held-out-scene confusion matrix

True ↓ / Pred →	Tomato	Sugar	Master	Tuna	Bleach	Total
Tomato	74	0	0	31	45	150
Sugar	0	29	0	0	46	75
Master	7	0	68	0	0	75
Tuna	69	0	0	6	0	75
Bleach	32	12	0	20	11	75
Predicted totals	182	41	68	57	102	450

All 262 off-diagonal predictions are accounted for in the eight observed transitions below. A transition identifies an error pattern; it does not, by itself, establish a cause such as occlusion or mislabeled ground truth.

True → predicted	Count	% of true class	% of all 262 errors
tuna → tomato	69	92.00%	26.34%
sugar → bleach	46	61.33%	17.56%
tomato → bleach	45	30.00%	17.18%
bleach → tomato	32	42.67%	12.21%
tomato → tuna	31	20.67%	11.83%
bleach → tuna	20	26.67%	7.63%
bleach → sugar	12	16.00%	4.58%
master → tomato	7	9.33%	2.67%

6.3.1 Directionality of the largest prediction columns

The model predicted tomato_soup_can 182 times, but only 74 of those predictions were correct (precision 40.66%). Its 108 tomato false positives came from tuna_fish_can (69), bleach_cleanser (32), and master_chef_can (7). Conversely, 76 actual tomato objects were sent to bleach (45) or tuna (31). This two-way view distinguishes an over-predicted label from a frequently misclassified true class.

The bleach_cleanser prediction column contains 102 cases, of which only 11 are actually bleach (precision 10.78%): 46 were sugar and 45 were tomato. These observations describe which labels attract mistakes. They do not identify whether context, pose, or dataset composition caused them.

6.3.2 Concentration of errors

Three directed transitions — tuna → tomato (69), sugar → bleach (46), and tomato → bleach (45) — account for 160 of 262 errors (61.07%). Across true classes, tomato, tuna, and bleach contribute 209 of 262 mistakes (79.77%). This concentration gives the manual review a defined starting point without assuming that the largest raw-error class has the largest conditional error rate.

Diagnostic aggregation	Observed count	Share of 262 errors
Three largest directed confusions	160	61.07%
Tomato + tuna + bleach true-class errors	209	79.77%
Remaining true-class errors (sugar + master)	53	20.23%

6.4 Representative Visualizations and Notebook Outputs

Figures 1–6 below are actual chart outputs embedded in the supplied executed Week 05 notebook. Figures 7–8 are real Week 04 example grids reused in its Part C2. These are diagnostic visualizations of the recorded E1 evaluation, not additional model-training results.

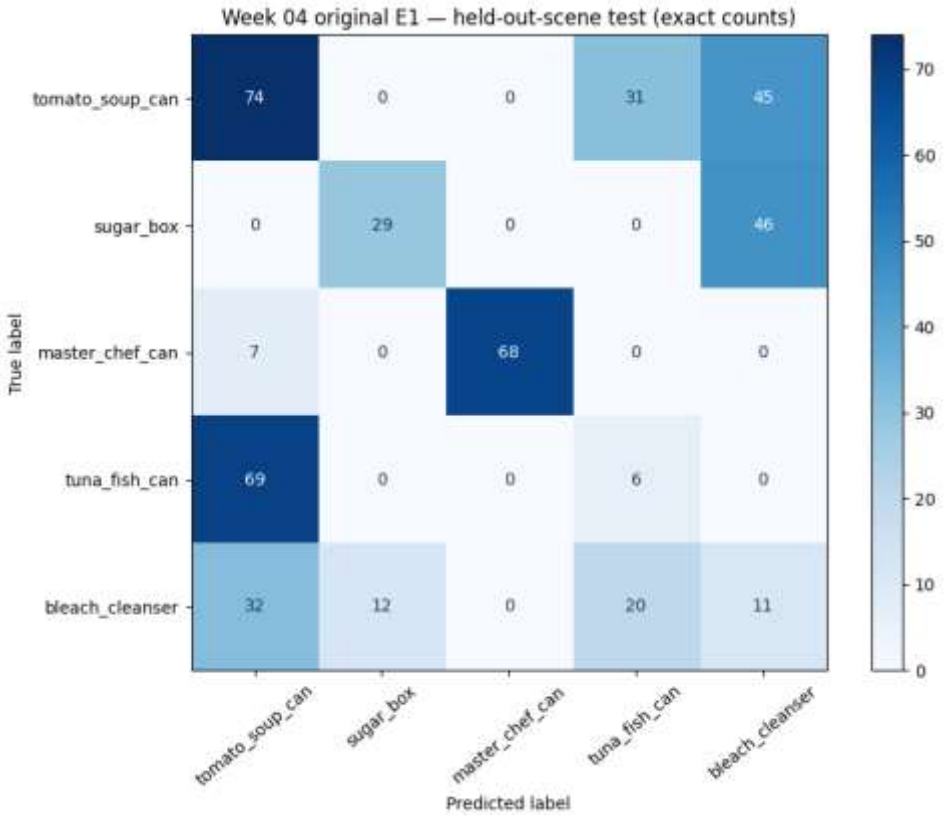


Figure 1. Complete E1 held-out-scene confusion matrix: all 450 samples, true classes as rows and predictions as columns (Week 05 Part B).

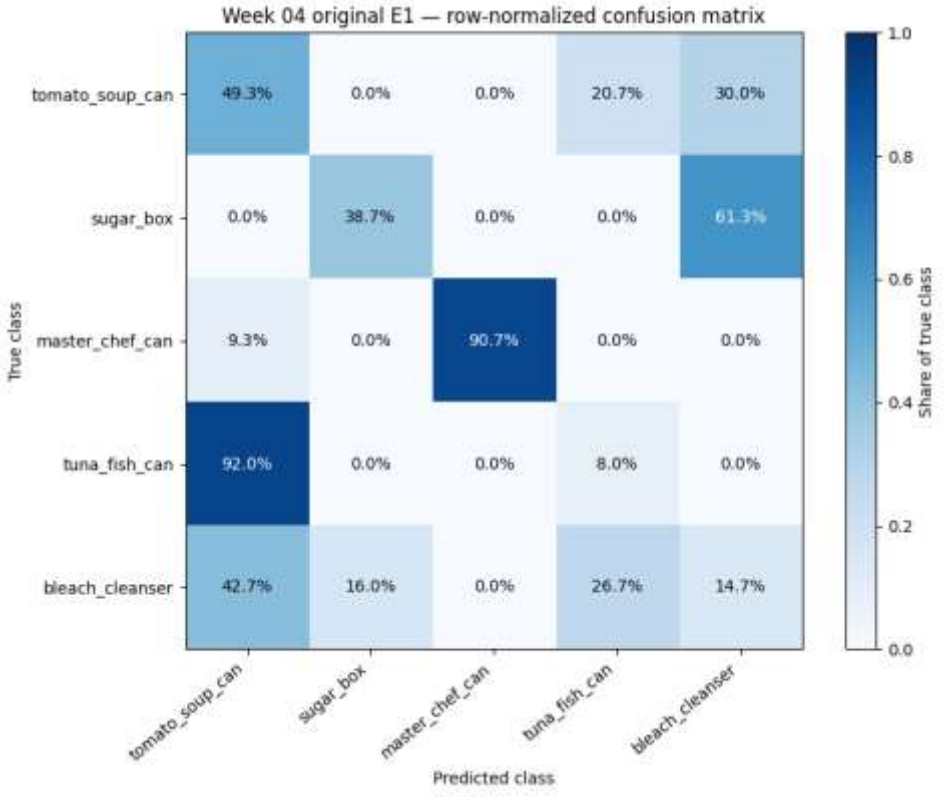


Figure 2. Row-normalized confusion matrix: the share of each true class mapped to every predicted class (Week 05 Part B).

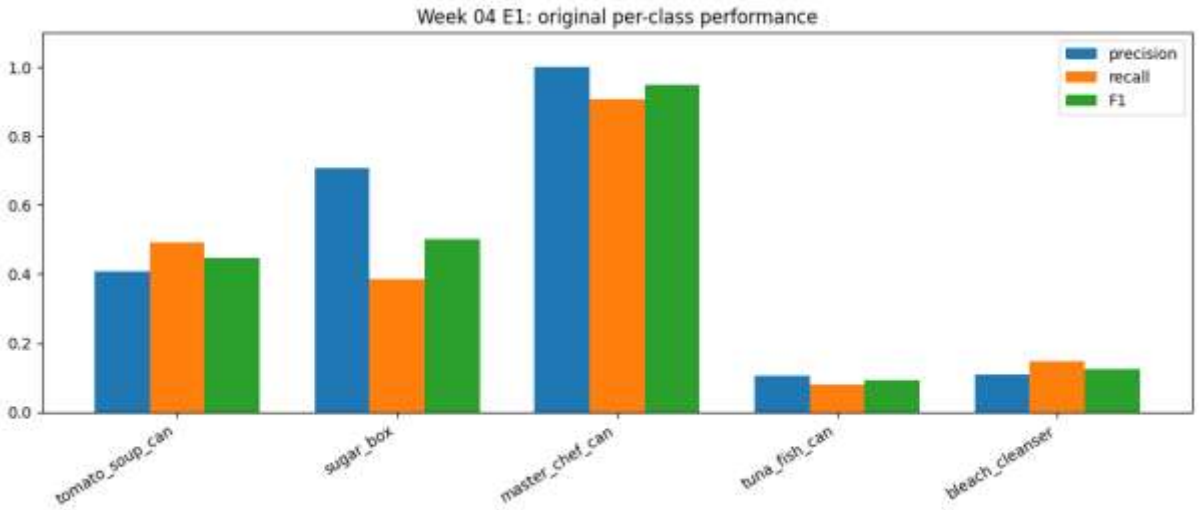


Figure 3. Precision, recall, and F1 by class; master_chef_can contrasts with the weak tuna_fish_can and bleach_cleanser results (Week 05 Part C).

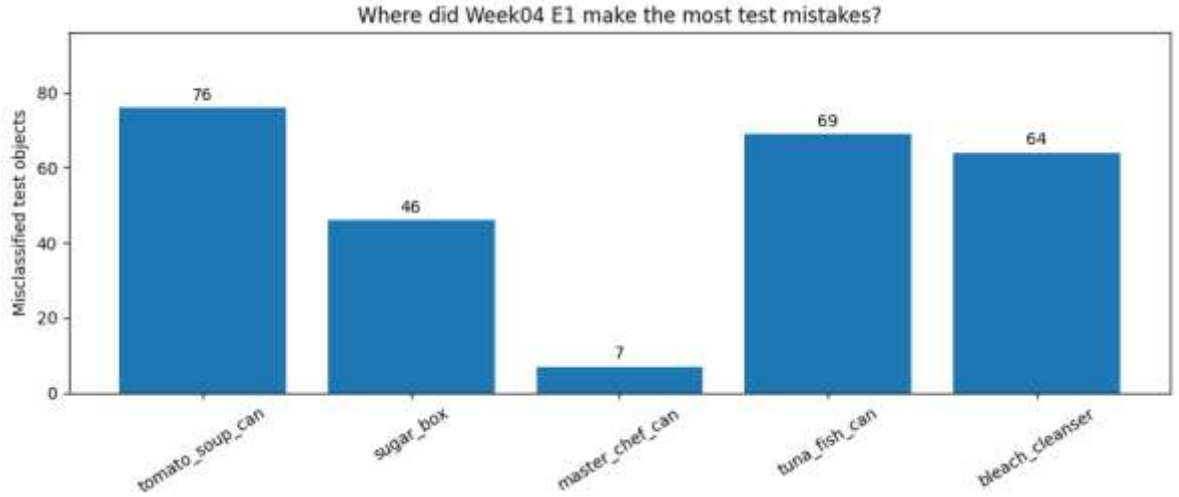


Figure 4. Absolute errors per true class: 76, 46, 7, 69, and 64 respectively (Week 05 Part C).

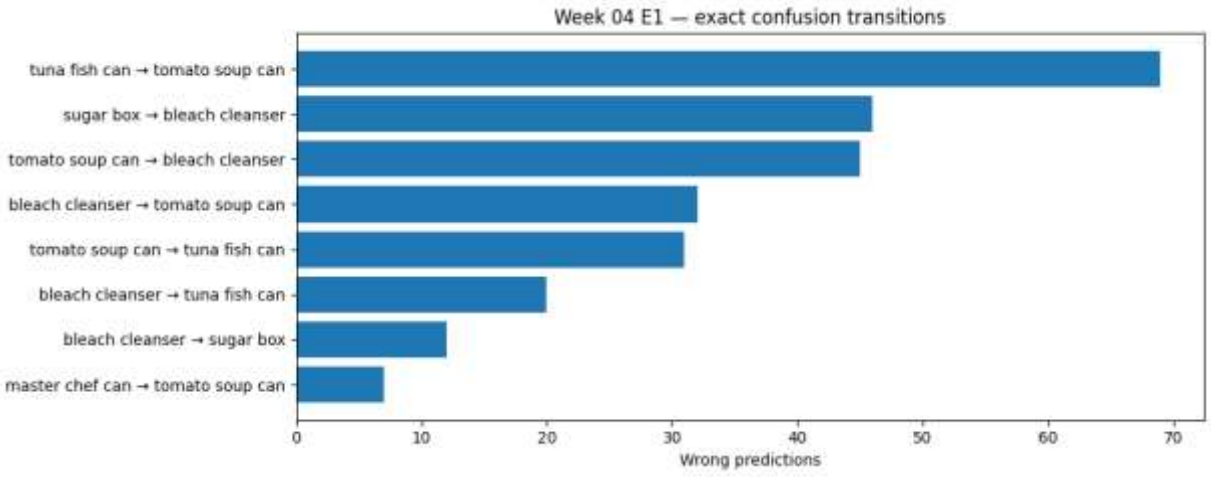


Figure 5. All eight nonzero error transitions, sorted by their observed frequencies (Week 05 Part C1).

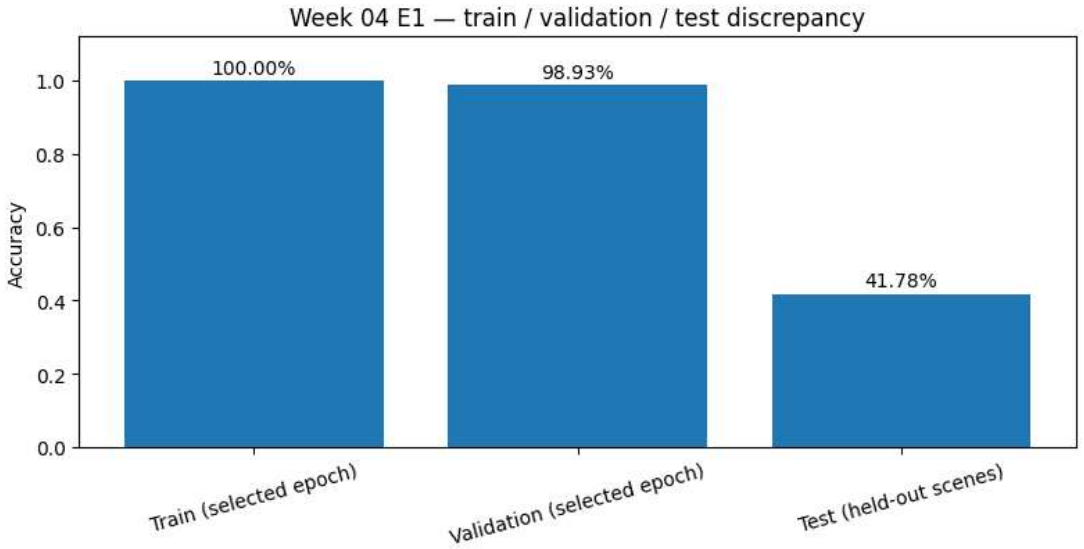


Figure 6. Original E1 accuracy gap across the selected-epoch training, validation, and held-out scene test sets (Week 05 Part C1).

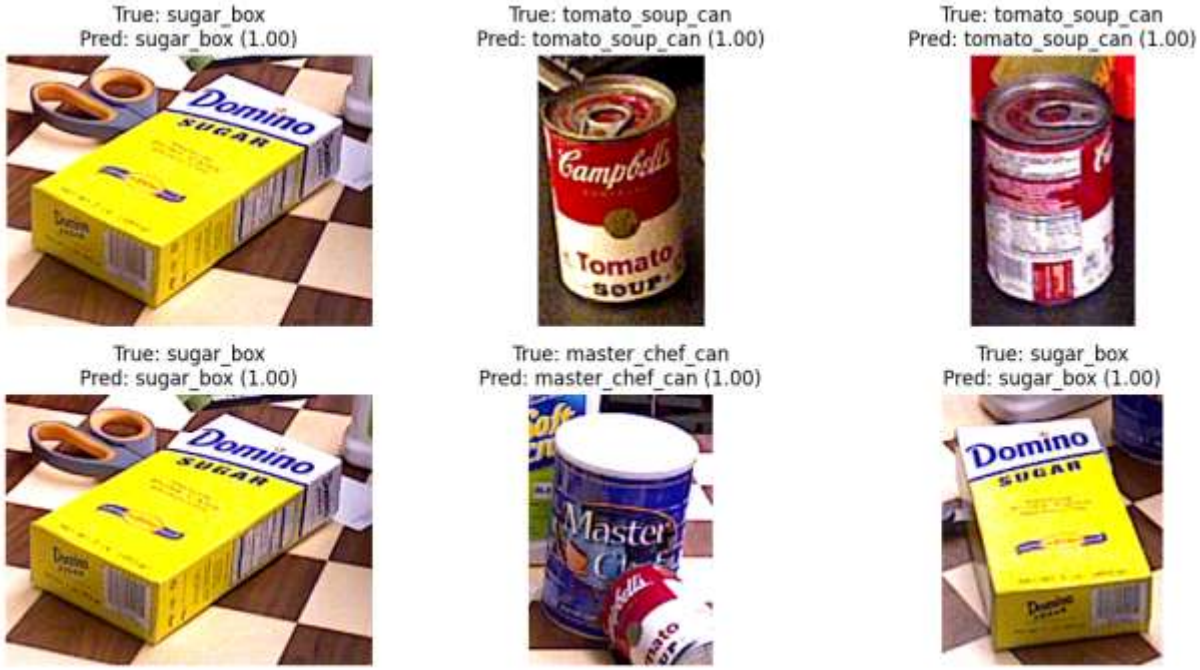


Figure 7. Correct original E1 prediction examples, preserved from the executed Week 04 notebook and displayed in Week 05 Part C2; confidence labels are the originals.



Figure 8. Incorrect original E1 examples, including highly confident mistakes and visible co-occurring objects in some crops. These examples are illustrative, not a random prevalence estimate.

6.5 Saved notebook artifacts and unavailable optional diagnostics

The executed Part F saved original_week04_E1_full_confusion_matrix.csv, week05_classwise_exact.csv, week05_all_error_transitions.csv, week05_observed_split_accuracy.csv, week05_issue_hypotheses_to_verify.csv, and manual_error_review_template.csv under outputs/week05. The manual-review file contains headers but zero reviewed examples because the original per-image prediction CSV was unavailable in this run.

Part D explicitly reported that final_test_predictions.csv was not found; it therefore did not produce scene-wise metrics, per-image confidence-bin tables, or newly reconstructed crops. Part E checkpoint recovery was not requested. The six plots and the old example grids above are the available, verified visual evidence; the missing diagnostics are not populated with guesses.

7. Failure Analysis and Interpretation

7.1 What the measurements establish

- Validation accuracy (98.93%) did not predict held-out-scene test accuracy (41.78%), despite the original E1 reaching 100.00% training accuracy. The observed validation–test gap is 57.15 percentage points.
- The principal directed confusion is tuna_fish_can → tomato_soup_can: 69 of 75 tuna samples (92.00%). Other major transitions are sugar → bleach (46) and tomato → bleach (45).
- Class behavior is highly uneven: master_chef_can recall was 90.67%, compared with 8.00% for tuna and 14.67% for bleach. Overall accuracy alone conceals this asymmetry.
- The original error examples include confidence values near 1.00, so a confident softmax output cannot be interpreted as reliable correctness on new scenes.

7.2 Candidate explanations requiring verification

Potential scene/viewpoint distribution mismatch, background and nearby-object contamination within padded crops, occlusion/scale changes, and annotation/crop problems are reasonable checks based on the split discrepancy and original example grids. However, this executed notebook did not measure their individual contributions. In particular, it did not verify any mislabeled sample, quantify per-scene performance, or count high-confidence failures across all 450 examples.

7.3 Prioritized diagnostic issues for Week 06

Priority	Evidence observed	Verification action
1 • Split representativeness	98.93% validation vs 41.78% test	Compare scene, viewpoint, lighting and co-occurring object distributions; preserve a genuinely untouched test split.

Priority	Evidence observed	Verification action
2 • Tuna/tomato confusion	69/75 tuna predicted as tomato	Inspect paired raw frames and GT crops; check pose, apparent size and background.
3 • Crop context	Some original wrong examples include nearby objects	Compare GT crop with frame context; manually tag contamination and occlusion in selected examples.
4 • Sugar/bleach and bleach spread	46 sugar → bleach; bleach errors to 3 classes	Review failure images across those transitions; inspect object/background overlap.
5 • Labels and boxes	Possible issue only; none confirmed	Check source BOP scene_gt.json and scene_gt_info.json only where inspected examples are suspicious.

Recommended method: first recover the existing original Week 04 E1 predictions/checkpoint and original scene RGBs without retraining; then use the same error-review template to tag a defined sample. Turn verified tags into one-variable-at-a-time Week 06 experiments. Do not tune directly to the held-out test scenes.

8. Conclusion and Next Step

Week 05 delivered a full, exact class-wise error analysis of the selected Week 04 E1 classifier using the same lightweight YCB-V BOP'19 subset. The matrix accounts for all 450 test predictions and 262 mistakes, and the computed metrics expose strong class-specific failures plus a 57.15-point validation–test discrepancy. Diagnostic plots, the eight directed confusion transitions, and original correct/incorrect example grids document these results without inventing new training outcomes.

The most important next step is to recover the Week 04 final_test_predictions.csv (or the original E1 checkpoint plus matching images) and perform a structured visual audit. This will test the scene-mismatch and crop-contamination hypotheses before Week 06 changes the model or data pipeline. The present report distinguishes observed results, qualitative indications, and unverified explanations.